

AI Fraud-Spike Detector

Project Concept, System Context & Blueprint Foundation

A foundational context document for an AI/ML defense-only risk monitoring project

Purpose

This document captures the project's established concept, terminology, conceptual workflow, evaluation requirements, possible technical directions, and open design questions. It is intended to serve as context for a future AI assistant that will help turn the concept into a detailed technical blueprint.

Status: Foundational concept — implementation architecture and final modeling choices are intentionally not yet fixed.

1. Project Overview

Project name: AI Fraud-Spike Detector

Core objective: Build a defense-only AI/ML system that detects unusual increases (“spikes”) in suspicious/fraudulent transaction activity over time and alerts a merchant/business so that appropriate defensive investigation or response can be initiated.

Central distinction: This is not simply an individual transaction fraud detector. The project focuses on abnormal changes in the overall pattern of suspicious activity across transactions over time.

What the system is trying to answer

“Is there a sudden and statistically meaningful increase in suspicious activity compared with what is normally expected?”

The project is motivated by the fact that businesses may continuously receive thousands of transactions and may already experience some baseline level of suspicious or fraudulent activity. A useful spike detector therefore needs to distinguish normal variation from a potentially significant change.

- Potential consequences of a genuine spike include increased financial losses, coordinated fraudulent activity, attacks targeting a merchant/payment system, abnormal behavior across accounts/devices/transactions, and increased operational burden on fraud/risk teams.
- The project remains strictly defense-only.

2. The Problem

A merchant may process thousands of transactions continuously. Some transactions may eventually be confirmed as fraudulent, while others are legitimate. The presence of some fraud does not automatically mean the merchant is experiencing a major fraud event.

For example, a historical period could contain 10,000 transactions with approximately 100 confirmed fraudulent transactions, corresponding to an approximately 1% fraud rate. That level may form part of the merchant's normal baseline.

The important question is not simply “does fraud exist?” but whether the level or pattern of suspicious activity has changed abnormally.

A genuine spike could indicate a coordinated event, an emerging attack, a merchant-specific problem, or another unusual behavioral shift that deserves attention.

The detector is therefore intended as a risk-monitoring layer that can alert the business when suspicious activity becomes unusually concentrated or elevated.

3. Individual Fraud Detection vs Fraud-Spike Detection

Individual Fraud Detector

Question: “Is this particular transaction suspicious or potentially fraudulent?”

- Example signals may include unusually high amount, a new device, unusual location, or unusual customer behavior.
- An individual model might output a risk score such as 94%.
- A risk score is not a confirmed fraud verdict. It means the transaction may deserve review or additional verification according to the business's risk policy.

Fraud-Spike Detector

Question: “Is there an unusual increase in suspicious activity across many transactions during a particular period?”

Suppose normal suspicious activity is around 0.8%, 1.0%, 1.1%, 0.9%, and 1.0%, followed by 8.0%. The detector should recognize that the overall pattern has changed dramatically.

The Fraud-Spike Detector can therefore act as an additional risk-monitoring layer rather than necessarily replacing an individual transaction fraud model.

4. The Role of Historical Fraud Labels

AI does not magically know which transactions are fraudulent. When supervised learning is used, historical transaction data needs outcomes that can serve as labels.

Transaction	Outcome
#1001	Legitimate
#1002	Legitimate
#1003	Confirmed fraud
#1004	Legitimate
#1005	Confirmed fraud

Conceptual sources of labels can include customer reports, bank/payment-provider investigation, confirmed disputes or chargebacks, merchant investigation, or other legitimate confirmation processes.

Ground truth vs model prediction

Ground truth: what was eventually confirmed to be fraudulent or legitimate.

Model prediction: what the AI predicts or risk-scores before the final outcome is known.

This distinction is essential because a model can be wrong. A high risk score is not proof of fraud.

- True Positive: the system flags something that is later confirmed as fraud.
- False Positive: the system flags something that is later found to be legitimate.
- False Negative: fraud occurs but the system fails to flag it.
- True Negative: legitimate activity is correctly left unflagged, where this definition is appropriate.

The project should not assume that every available dataset has perfect labels. Label quality is an open research/design question.

5. The 10,000 Transaction Example

Suppose 10,000 transactions occur. After eventual investigation, 1,000 are confirmed fraudulent. This is historical ground truth; the real-time system does not know this number at the moment the transactions occur.

During operation, an individual transaction-risk component might assign risk levels such as:

- 7,500 → low risk
- 1,800 → medium risk
- 700 → high risk

The system does not automatically declare all 700 high-risk transactions fraudulent. They are flagged/risk-scored according to the system's eventual design.

Later investigation could reveal that 500 of the 700 flagged transactions were actually fraudulent and 200 were legitimate.

Outcome	Count
True Positives	500
False Positives	200
Actual fraudulent transactions not flagged (False Negatives)	300

This example illustrates why the system needs honest evaluation rather than simply reporting how many alerts it generated.

6. What “Aggregating Over Time” Means

Aggregation means grouping individual transactions into a defined time window and calculating summary statistics. The system first evaluates transactions individually or obtains transaction-level risk/suspicion signals, then summarizes those signals over time.

Example: 3:00–4:00 PM

- Total transactions = 1,000
- AI-flagged suspicious transactions = 20
- Suspicious rate = $20 / 1,000 = 2\%$

The same calculation can be repeated for each hour:

Time	Total Transactions	Suspicious Transactions	Suspicious Rate
10 AM	1,000	10	1.0%
11 AM	1,100	11	1.0%
12 PM	950	9	0.95%
1 PM	1,200	12	1.0%
2 PM	1,050	11	1.05%
3 PM	1,300	104	8.0%

Purpose: aggregation makes it possible to observe the overall behavior of suspicious activity rather than looking only at isolated transactions.

7. Why Normal Hours Matter

The normal-looking hours are not ignored. They are essential because they establish the baseline.

If suspicious rates repeatedly appear around 1.0%, 1.0%, 0.95%, 1.0%, and 1.05%, the system learns that suspicious activity normally stays around approximately 1% for that context.

If the observed rate then becomes 8.0%, the system has evidence of a large deviation from the established behavior.

We cannot identify a spike unless we understand what normal behavior looks like.

Normal does not mean zero fraud. A baseline can contain some amount of fraud. The detector is interested in whether the level or pattern changes abnormally.

8. What a “Spike” Actually Means

A spike should not be defined using an arbitrary fixed percentage such as “anything above 5%.” Conceptually, a spike means a significant deviation from expected or baseline behavior.

For example, a baseline near 1% followed by an observed rate near 8% may represent a major spike. But whether a change from 1% to 2% is significant depends on the context.

- Natural variability
- Merchant characteristics
- Time of day
- Day of week
- Seasonality
- Transaction volume
- Historical behavior
- Statistical uncertainty

Therefore, the exact definition of a spike is a future design/research decision.

9. Business Actions and Risk Levels

The system should not necessarily trigger the same response for every suspicious event. Conceptually, the business could have different levels of response.

Level	Meaning	Conceptual response
Low / normal	Behavior is within expected range.	No special spike response; normal fraud controls continue.
Mild anomaly	Behavior is somewhat unusual.	Monitor and/or investigate.
Strong anomaly	Behavior is substantially outside expected range.	High-priority alert and defensive investigation/action.

Illustrative alert

☐ FRAUD SPIKE DETECTED

Current suspicious rate: 8.4%

Expected baseline: ~1.1%

Suspicious activity: approximately 7.6× baseline

Affected period: 3:00–4:00 PM

Recommended response: investigate and apply appropriate defensive controls.

These numbers are illustrative only and must not be hard-coded as final thresholds.

10. False-Positive Cost

The original challenge explicitly requires honest metrics including false-positive cost.

False positives matter because unnecessary alerts can consume fraud-team time, inconvenience legitimate customers, cause unnecessary transaction review, and reduce trust in the system.

The system should therefore not optimize only for catching the maximum number of suspicious events. It should balance fraud losses avoided against the cost of incorrect alerts/interventions.

The exact financial and operational cost model remains a future design decision.

11. Evaluation

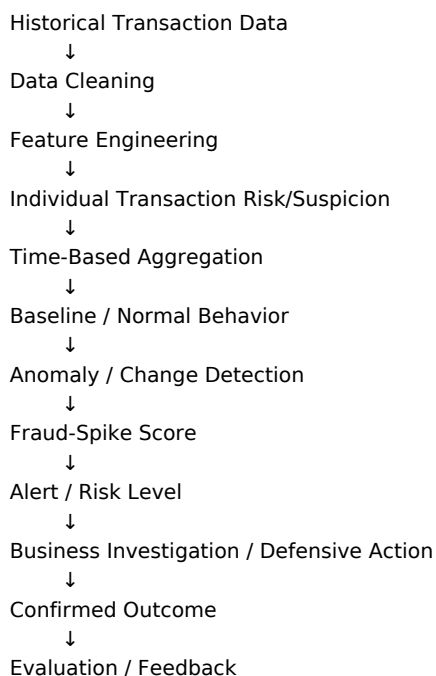
The project must use a held-out test set. Test data must not be used to train or tune the final model.

Where appropriate, the final evaluation should report:

- Precision
- Recall
- F1-score
- False-positive rate
- Confusion matrix
- False-positive cost
- False-negative cost, if appropriate
- Detection latency / time-to-detection
- Number of alerts
- Expected financial loss/savings, if the dataset and cost assumptions support this

No target score should be invented in advance. The final evaluation protocol should be designed around the chosen dataset and exact problem formulation.

12. Possible System Pipeline



This is a conceptual pipeline. The exact model architecture, data flow, and implementation choices have not yet been finalized.

Stage meanings

- Data cleaning prepares transaction records for analysis.
- Feature engineering creates useful transaction and behavioral signals.
- Transaction risk/suspicion represents the individual-level signal available to the spike detector.
- Time-based aggregation summarizes activity over selected windows.
- Baseline estimation characterizes expected behavior.
- Anomaly/change detection identifies deviations from expected behavior.
- The spike score summarizes the strength of the detected event.
- Alerting communicates the event and its context to the business.
- Investigation produces eventual outcomes that can support evaluation and, where appropriate, future model improvement.

13. Possible Technical Approaches to Investigate

No final model is selected at this stage. The following are candidate approaches to research.

Statistical methods

- Moving averages
- Rolling statistics
- Z-score
- IQR
- Control charts
- EWMA
- CUSUM

Time-series methods

- Trend analysis
- Seasonality
- Change-point detection
- Time-series anomaly detection

Machine-learning methods

- Isolation Forest
- One-Class SVM
- Local Outlier Factor
- XGBoost / LightGBM where supervised prediction is appropriate
- Other appropriate anomaly-detection methods

Potential advanced approaches

- Autoencoders
- Temporal neural networks
- Transformers for time-series
- Graph-based anomaly detection
- Bayesian change-point detection

Principle: start with strong statistical baselines before using complex deep-learning models. Advanced AI should be justified by the data and problem rather than added merely for the sake of using AI.

14. Supervised vs Unsupervised vs Hybrid

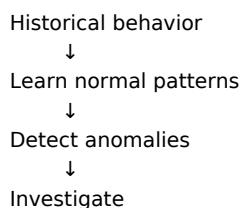
Supervised

Historical confirmed fraud labels exist.

Historical transactions
↓
Fraud / legitimate labels
↓
Train model
↓
Predict future transaction risk
↓
Aggregate risk over time

Unsupervised

There are insufficient or no confirmed fraud labels.



Important: an anomaly does not automatically mean fraud.

Hybrid

A possible future formulation could combine supervised transaction risk, unsupervised anomaly detection, and time-series spike detection. This is a future direction, not a final decision.

15. Important Conceptual Distinctions

Concept	Meaning
Fraud $\neq$ anomaly	An unusual transaction or pattern may still be legitimate.
Risk score $\neq$ certainty	A 94% model score is not a confirmed fraud verdict.
Suspicious activity $\neq$ confirmed fraud	Fraud confirmation may happen later through investigation or other legitimate processes.
Normal $\neq$ zero fraud	A baseline can contain some amount of fraud.
Spike $\neq$ arbitrary threshold	A spike should be determined relative to expected behavior and variability.

16. What the Project Is Not

The project is strictly defense-only. It is not:

- An offensive fraud tool
- A system for committing fraud
- A system for evading fraud detection
- A system for exploiting payment systems
- A system that automatically declares every flagged transaction fraudulent
- Merely a generic chatbot
- Merely an anomaly-detection visualization

It is a defense-only AI/ML risk monitoring system for detecting unusual increases in suspicious transaction activity.

17. What Is Already Decided

Established

- ☐ Project direction: Fraud-Spike Detector
- ☐ Defense-only objective
- ☐ Focus on detecting spikes in suspicious/fraudulent activity
- ☐ Individual transaction detection and aggregate spike detection must be distinguished
- ☐ Historical confirmed outcomes are needed where supervised learning is used
- ☐ Aggregation over time is central to spike detection
- ☐ Normal activity establishes the baseline

- ☐ False positives matter
- ☐ Honest precision/recall and cost-based evaluation are required
- ☐ A held-out test set is required

18. What Still Needs to Be Decided

Open research/design questions

- ☐ Exact fraud/loss scenario
- ☐ Dataset
- ☐ Data source
- ☐ Dataset quality
- ☐ Label quality
- ☐ Time granularity
- ☐ Features
- ☐ Definition of suspicious activity
- ☐ Definition of a spike
- ☐ Baseline methodology
- ☐ Statistical methods
- ☐ ML models
- ☐ Threshold selection
- ☐ Cost model
- ☐ Alerting mechanism
- ☐ Dashboard/UI
- ☐ Real-time vs batch architecture
- ☐ Deployment approach
- ☐ Final evaluation protocol

19. Final One-Paragraph Project Definition

The AI Fraud-Spike Detector is a defense-only machine-learning risk monitoring system designed to identify statistically and operationally significant increases in suspicious transaction activity over time. The system uses historical transaction outcomes and/or behavioral patterns to establish normal activity, evaluates incoming transactions or their risk signals, aggregates those signals across appropriate time windows, detects deviations from the expected baseline, and generates alerts for potential fraud spikes. The system does not assume that every anomaly is fraud and does not automatically declare flagged transactions fraudulent. Its effectiveness must be demonstrated using a held-out test set, honest precision/recall measurements, false-positive analysis, and an appropriate financial/operational cost framework.

Document status: Foundational project context. This document deliberately leaves dataset selection, exact feature definitions, spike formulation, final model architecture, thresholds, cost assumptions, deployment architecture, and UI decisions open for subsequent research and blueprint development.